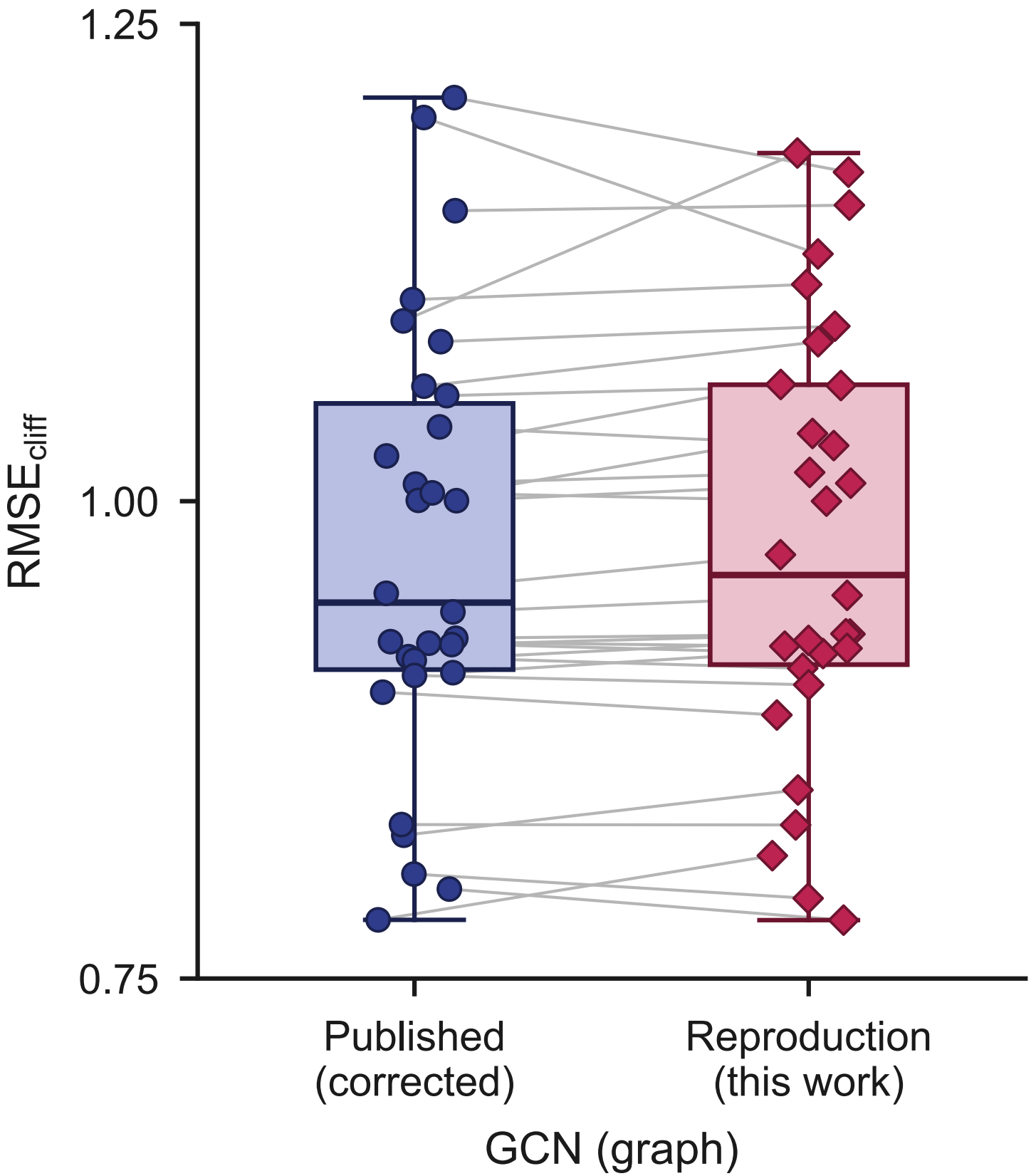
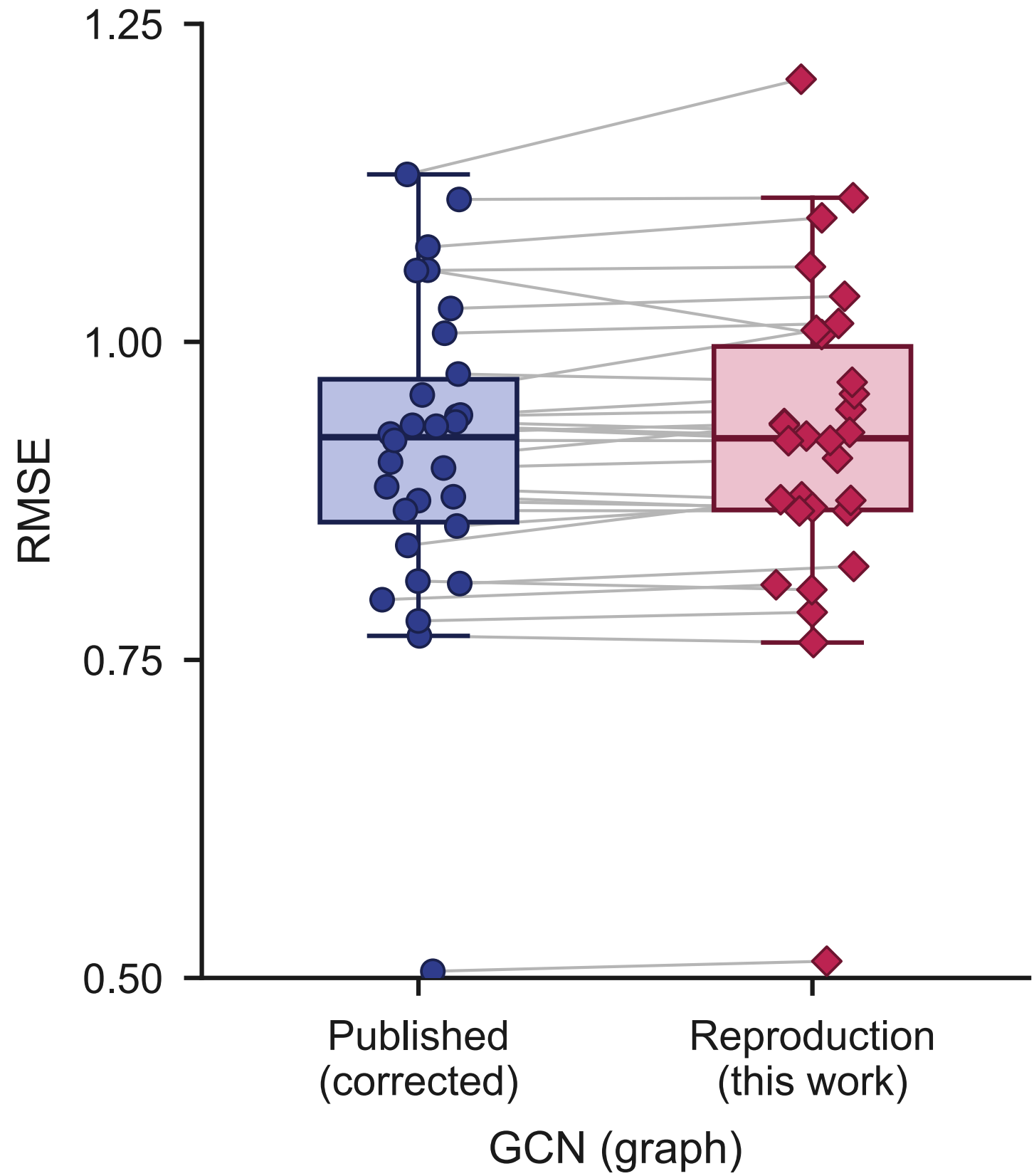
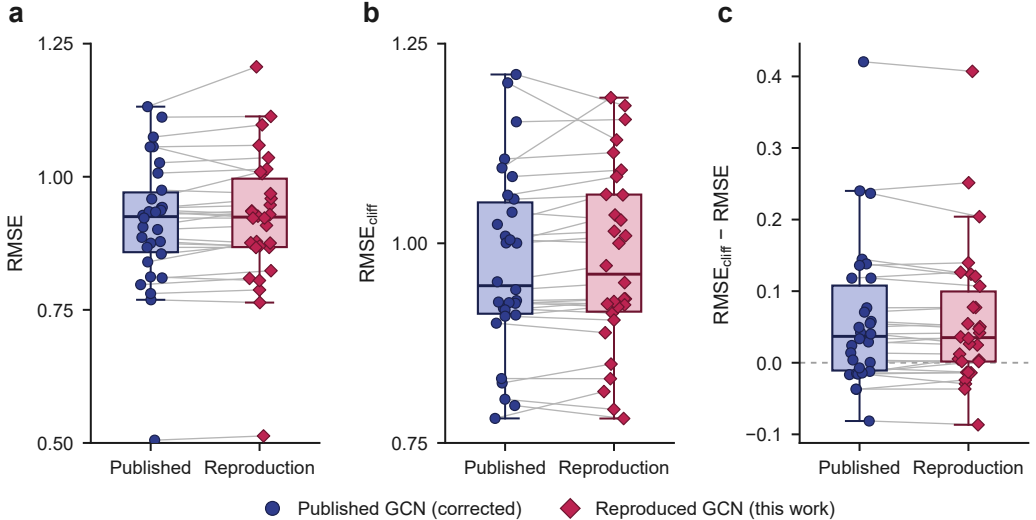
paired_cliff_rmse — Fig. 4a adaptation, paired GCN boxes (cliff)



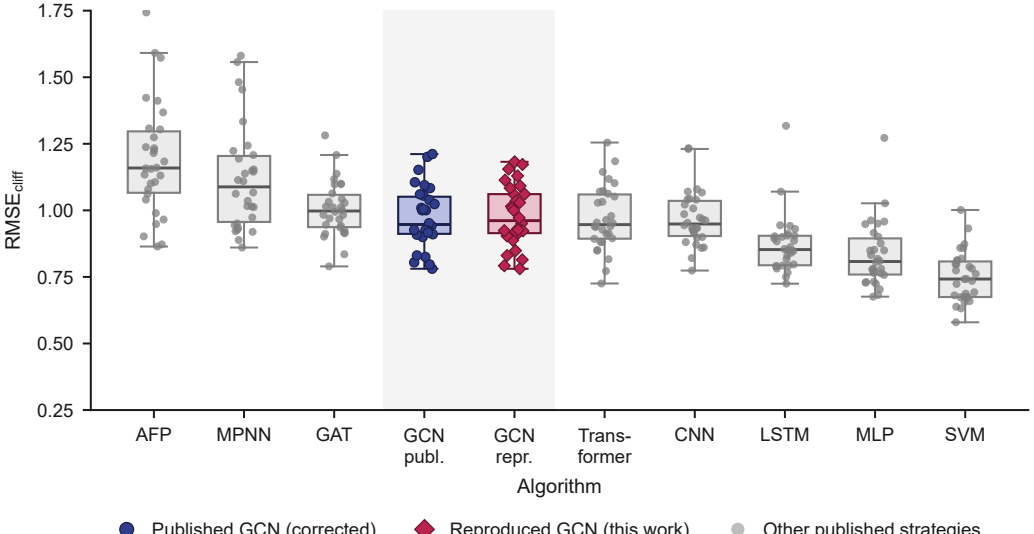
paired_rmse — Fig. 4a adaptation, paired GCN boxes (rmse)



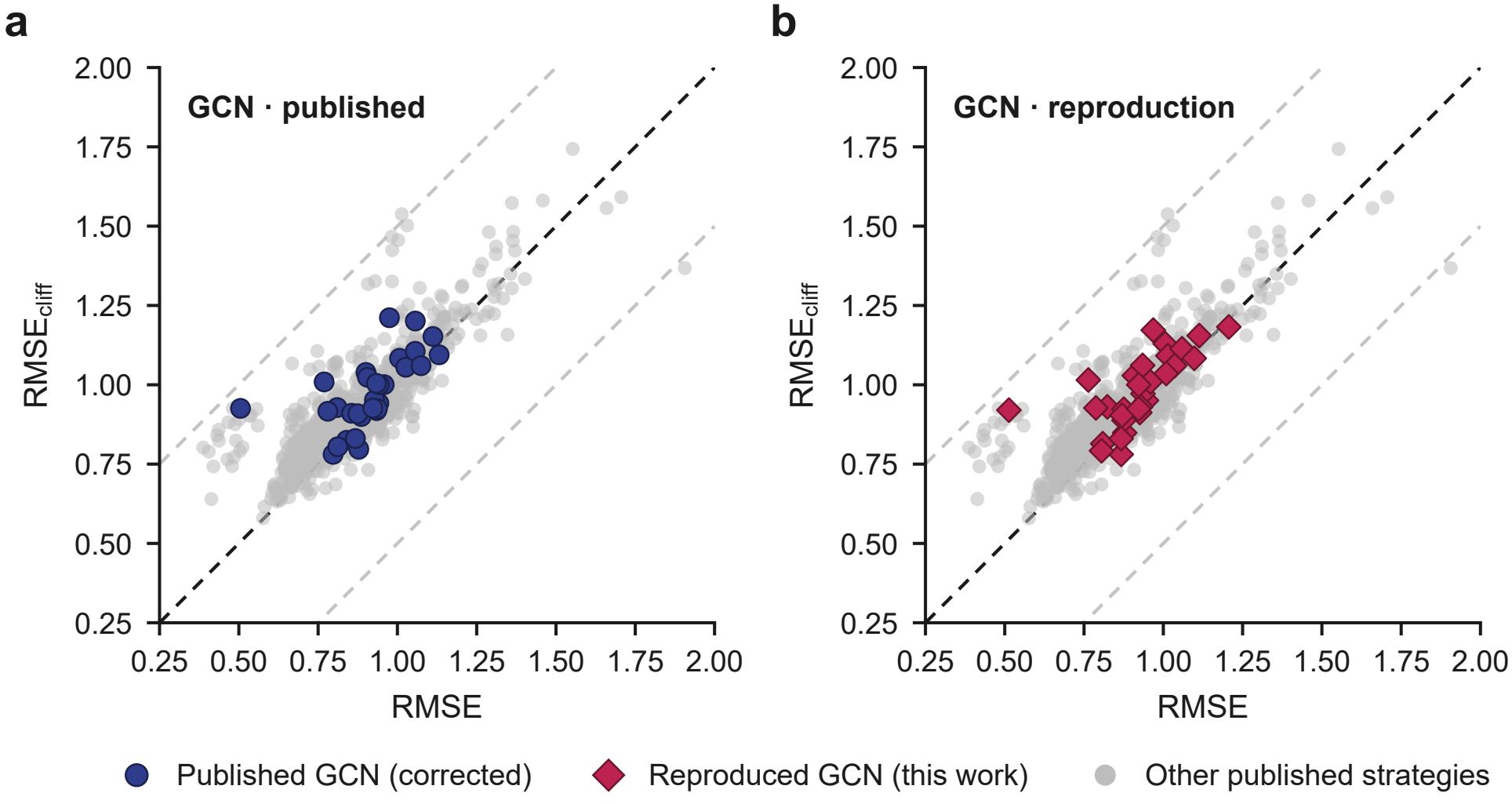
paired_error_metrics — Paired GCN boxes for RMSE, RMSE_cliff and their difference



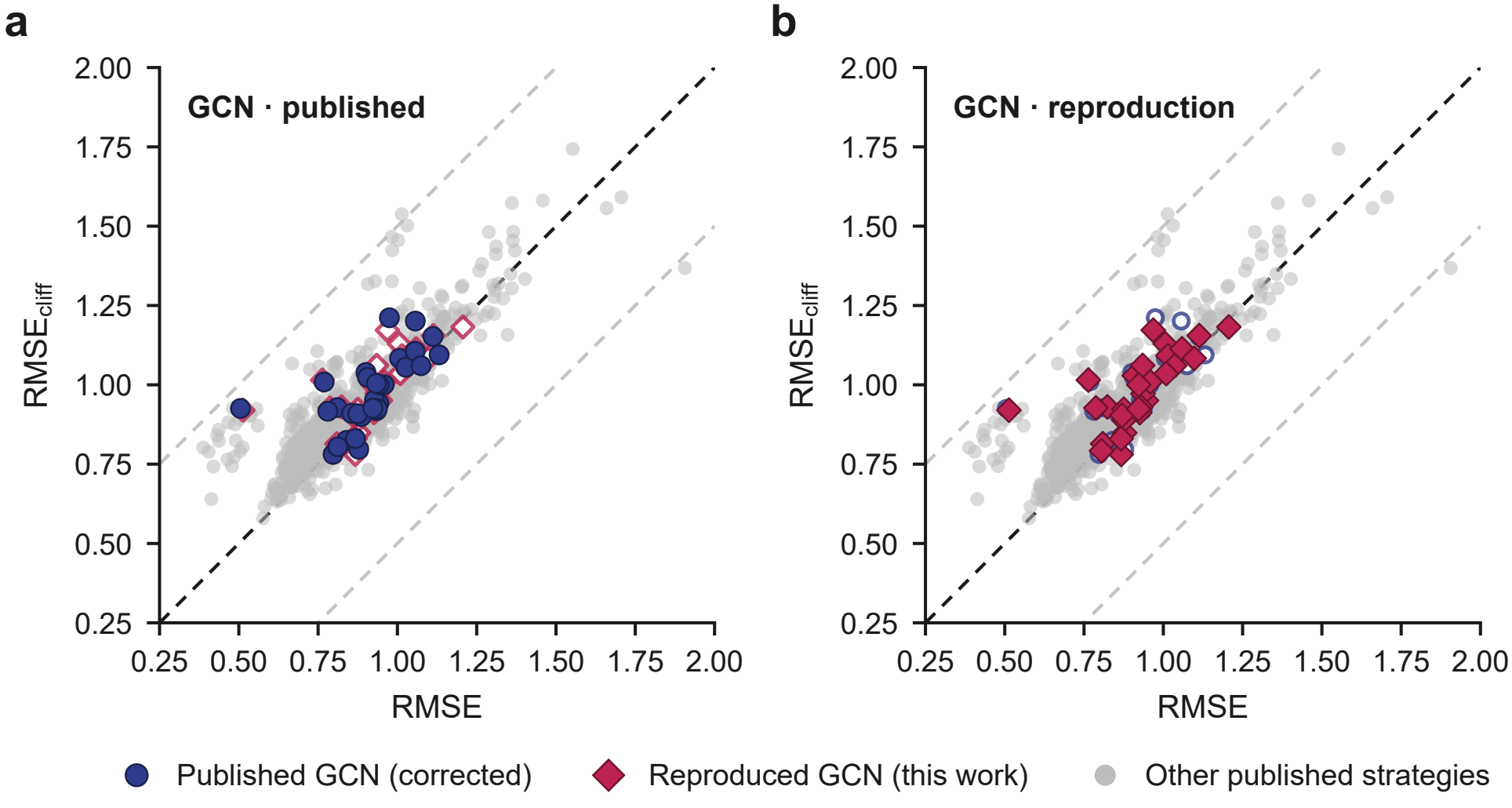
cliff_rmse_strategy_context — Fig. 4a with the reproduction inserted among published deep-learning strategies



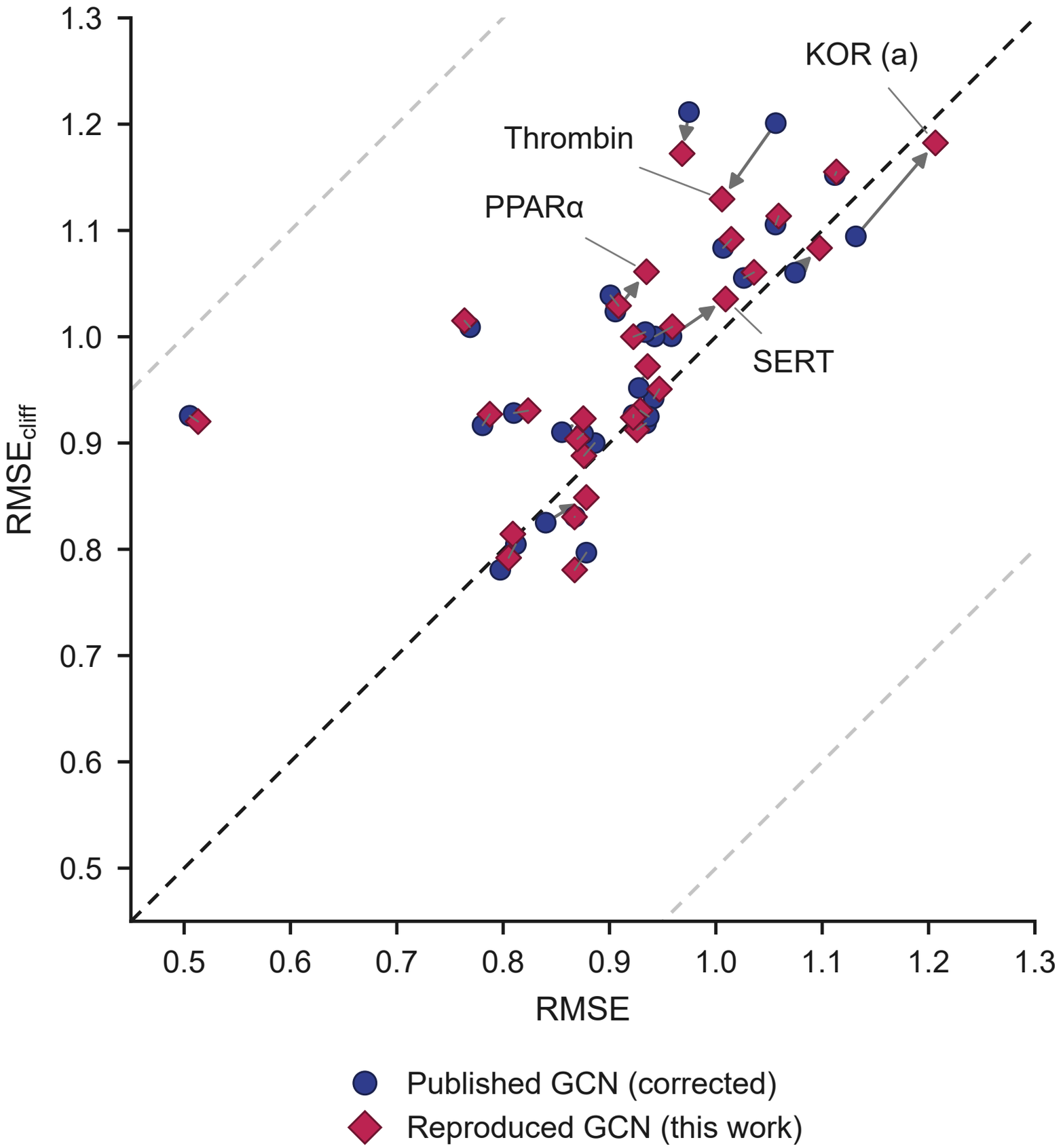
rmse_cliff_comparison — Fig. 4c: published and reproduced GCN in separate panels



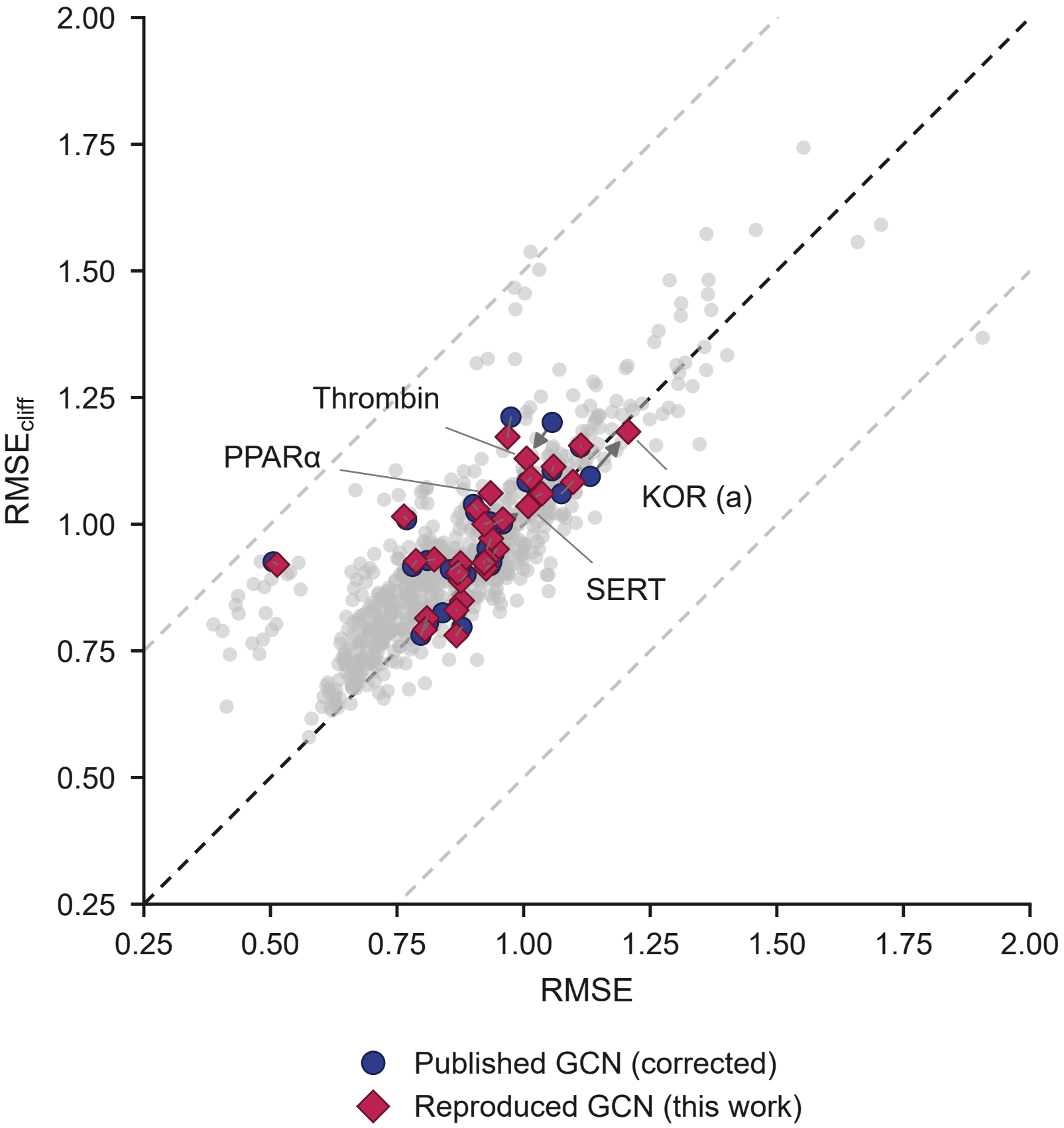
rmse_cliff_comparison_overlay — Fig. 4c: published and reproduced GCN in separate panels (with ghost)



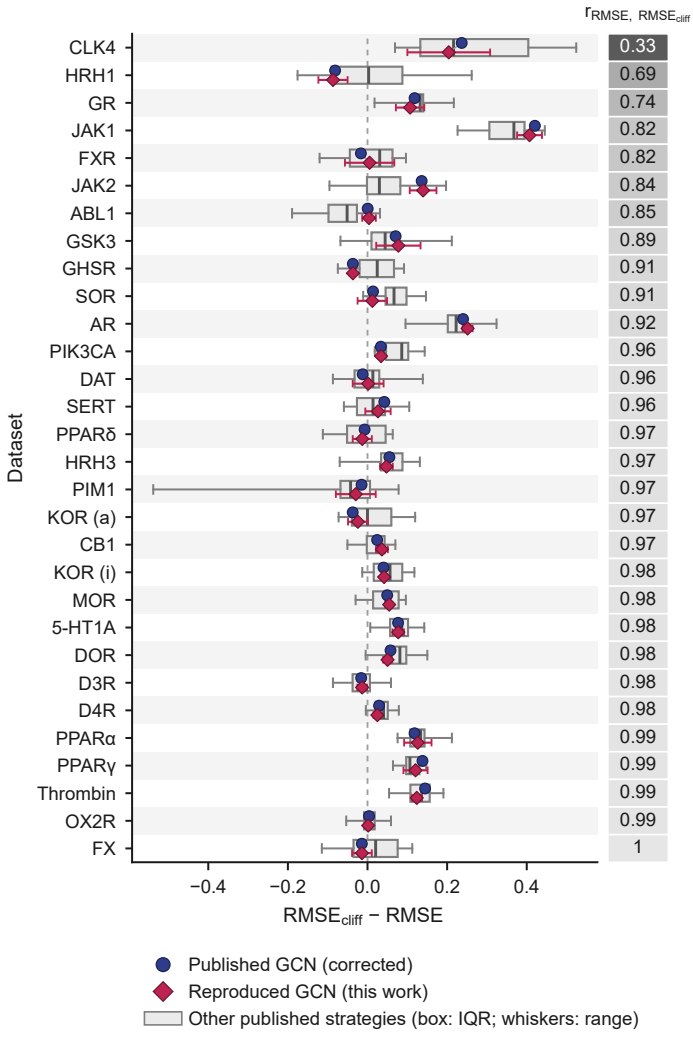
rmse_cliff_shifts_zoomed — Published-to-reproduced shift per dataset in the RMSE plane



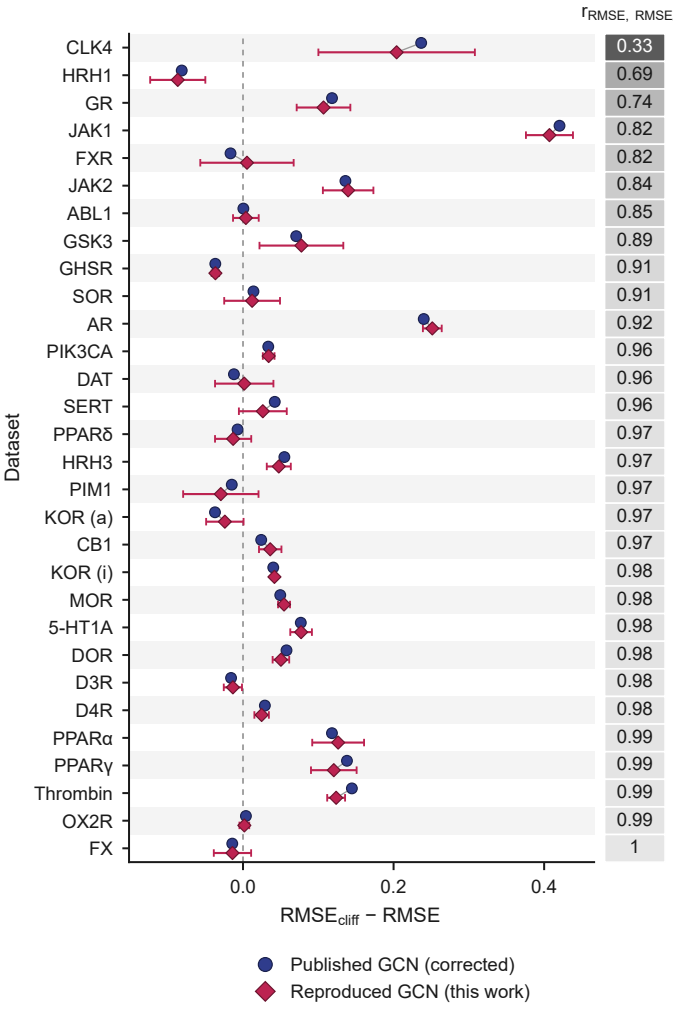
rmse_cliff_shifts_context — Published-to-reproduced shift per dataset in the RMSE plane



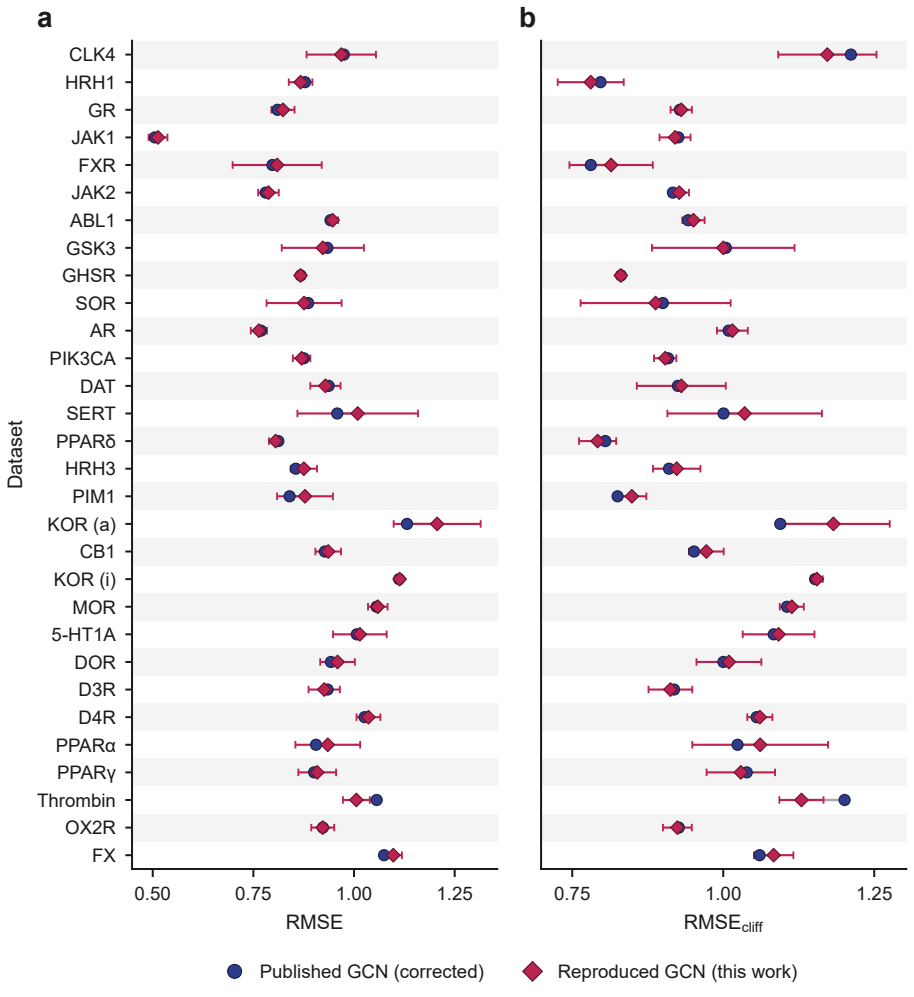
cliff_error_difference_strategy_context — Fig. 5a adaptation: RMSE_cliff − RMSE per dataset



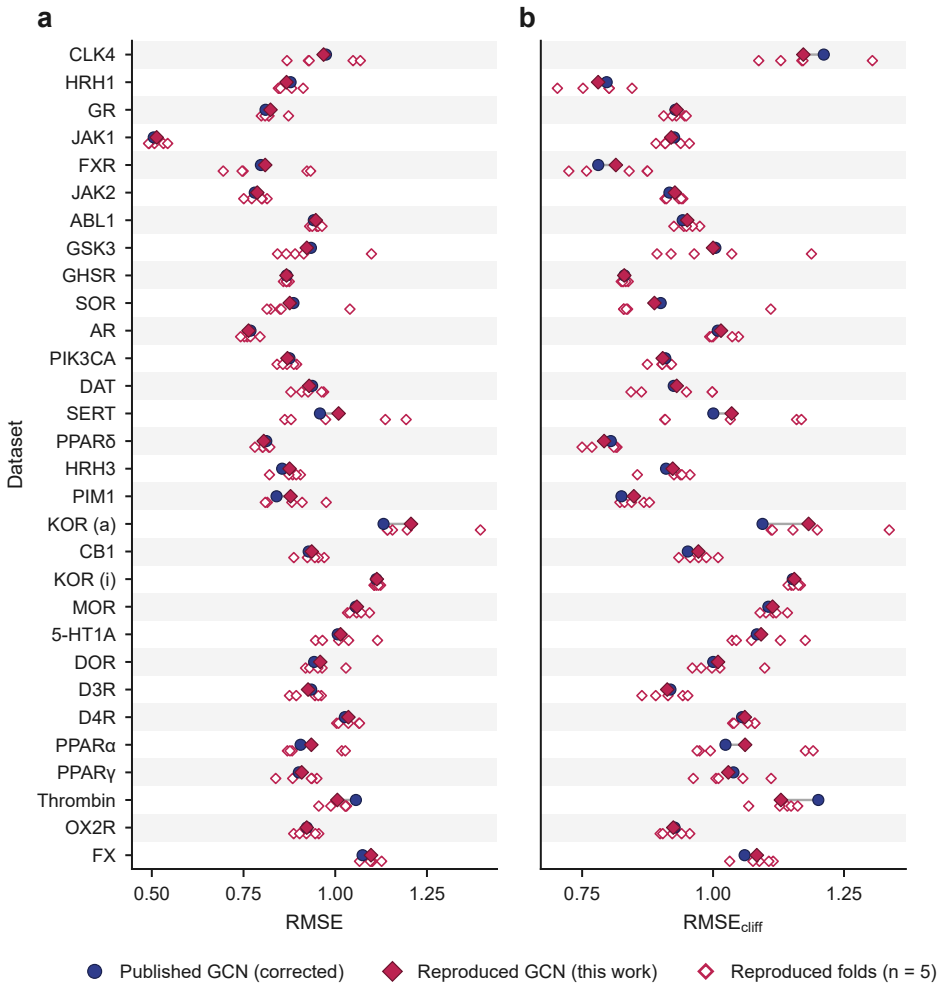
cliff_error_difference_by_dataset — Fig. 5a adaptation: RMSE_cliff − RMSE per dataset



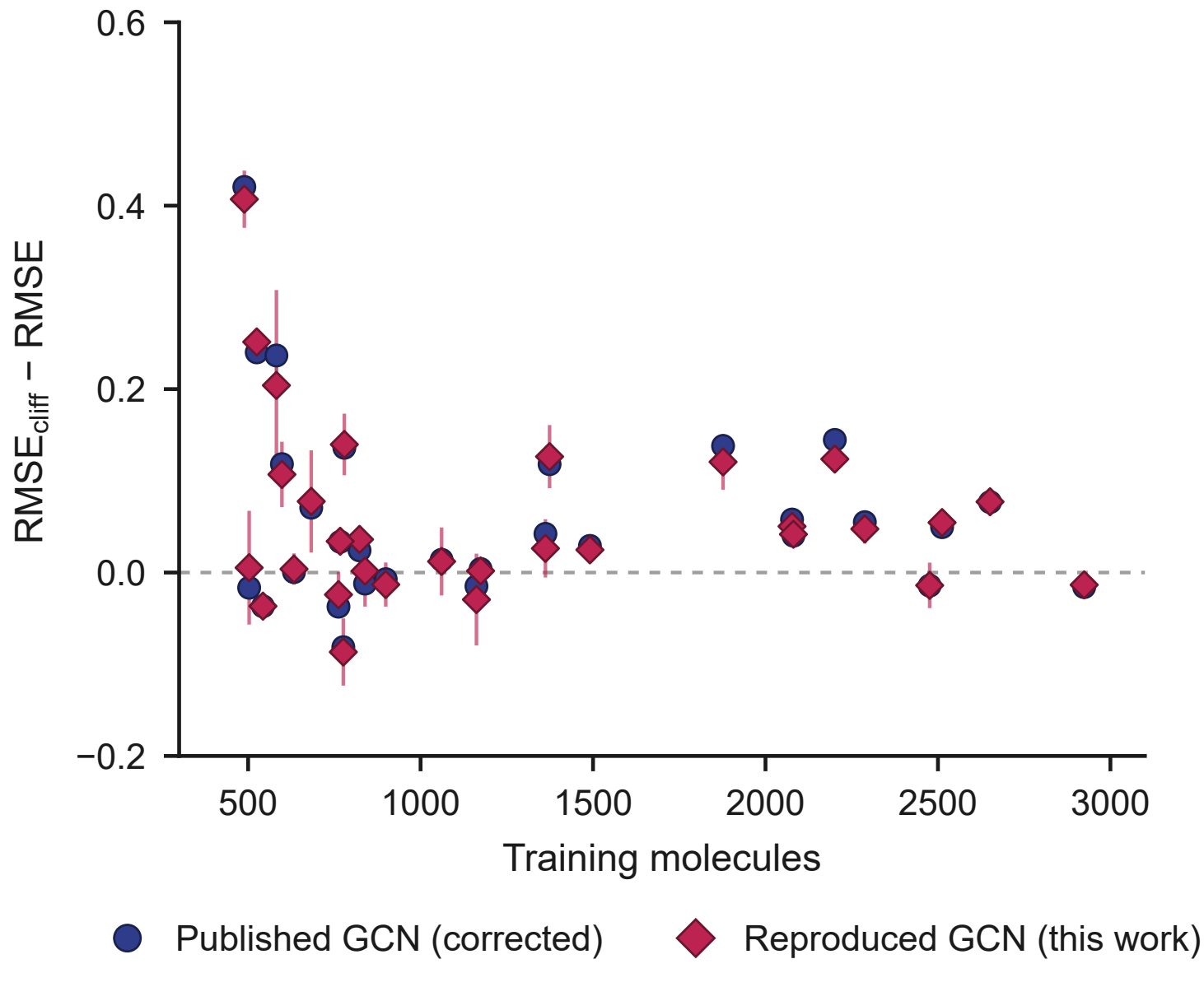
dataset_errors_mean_sd — Per-dataset published versus reproduced RMSE and RMSE_cliff



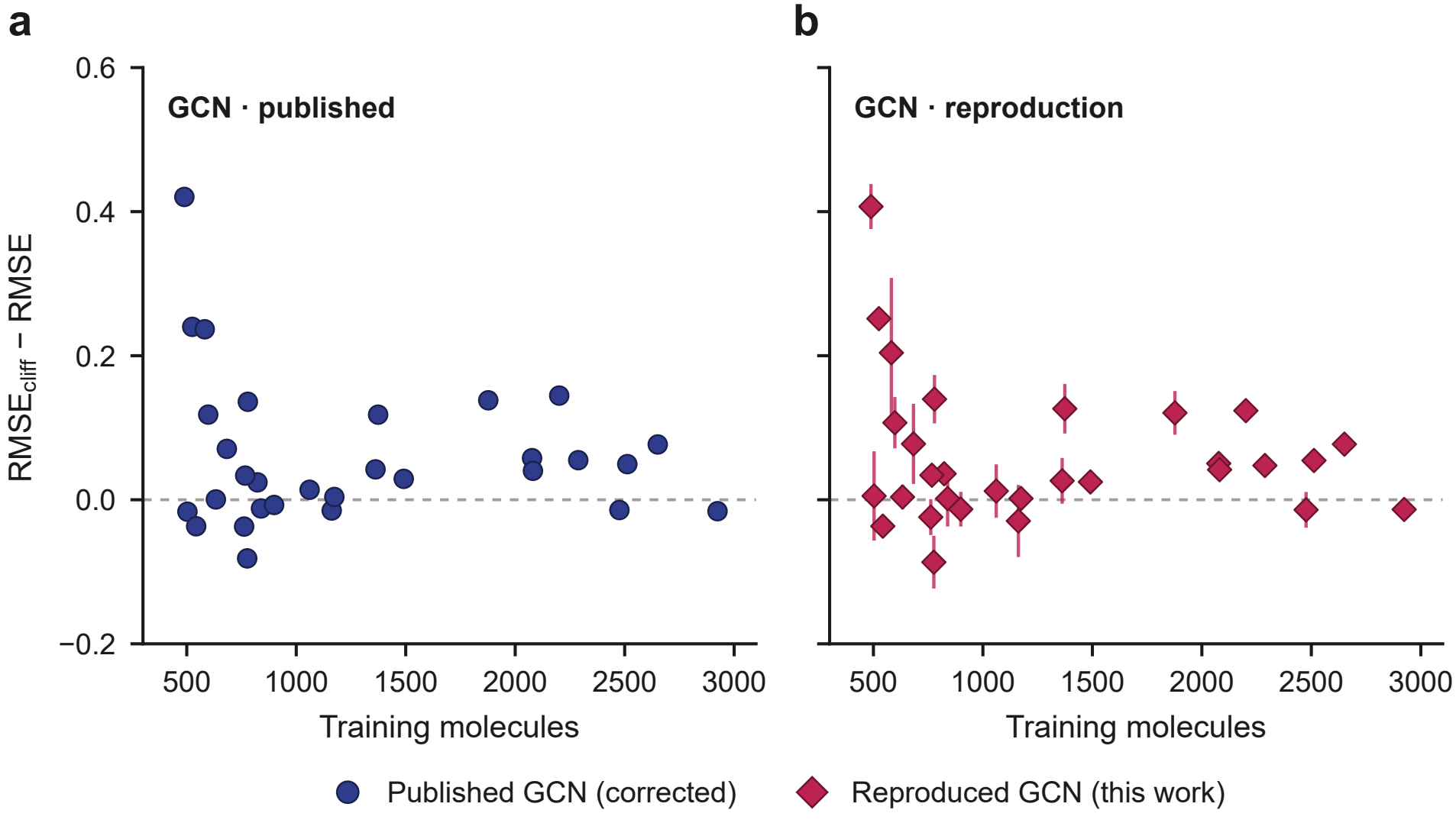
dataset_errors_by_fold — Per-dataset published versus reproduced RMSE and RMSE_cliff

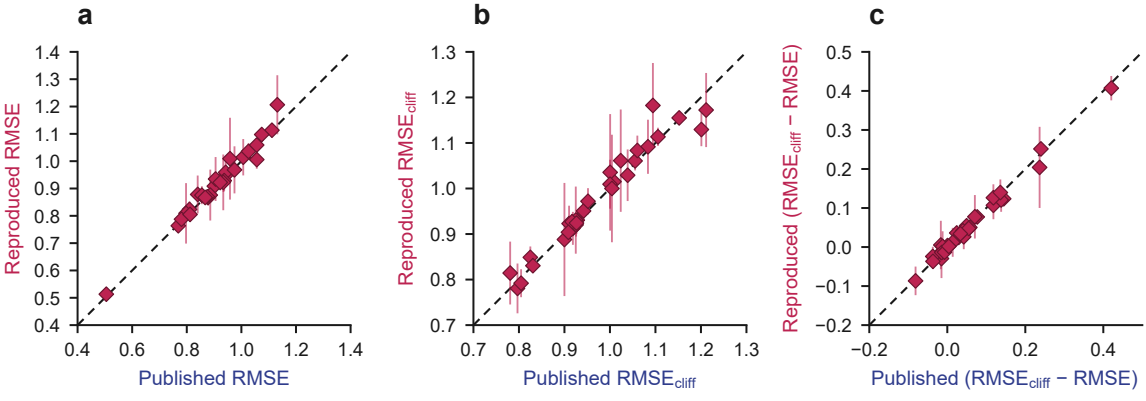


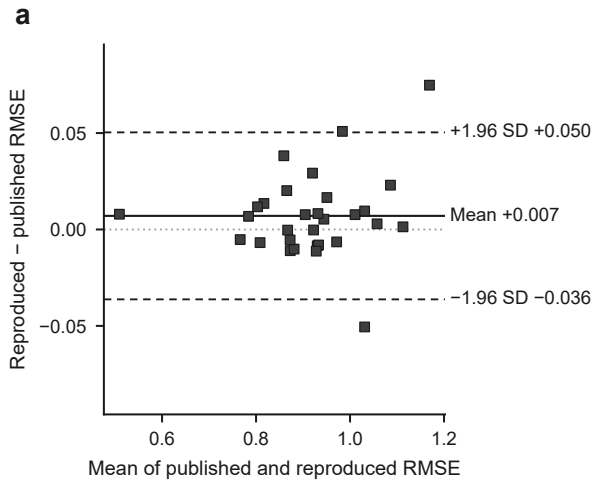
cliff_error_difference_training_size — Fig. 5d adaptation: RMSE_cliff − RMSE versus training-set size

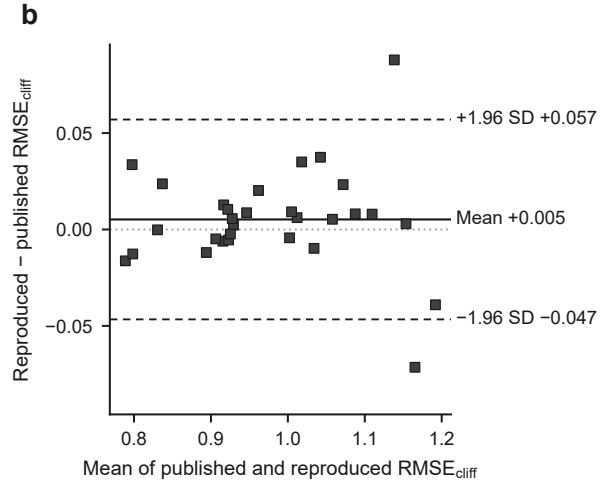


cliff_error_difference_training_size_comparison — Fig. 5d adaptation: RMSE_cliff − RMSE versus training-set size

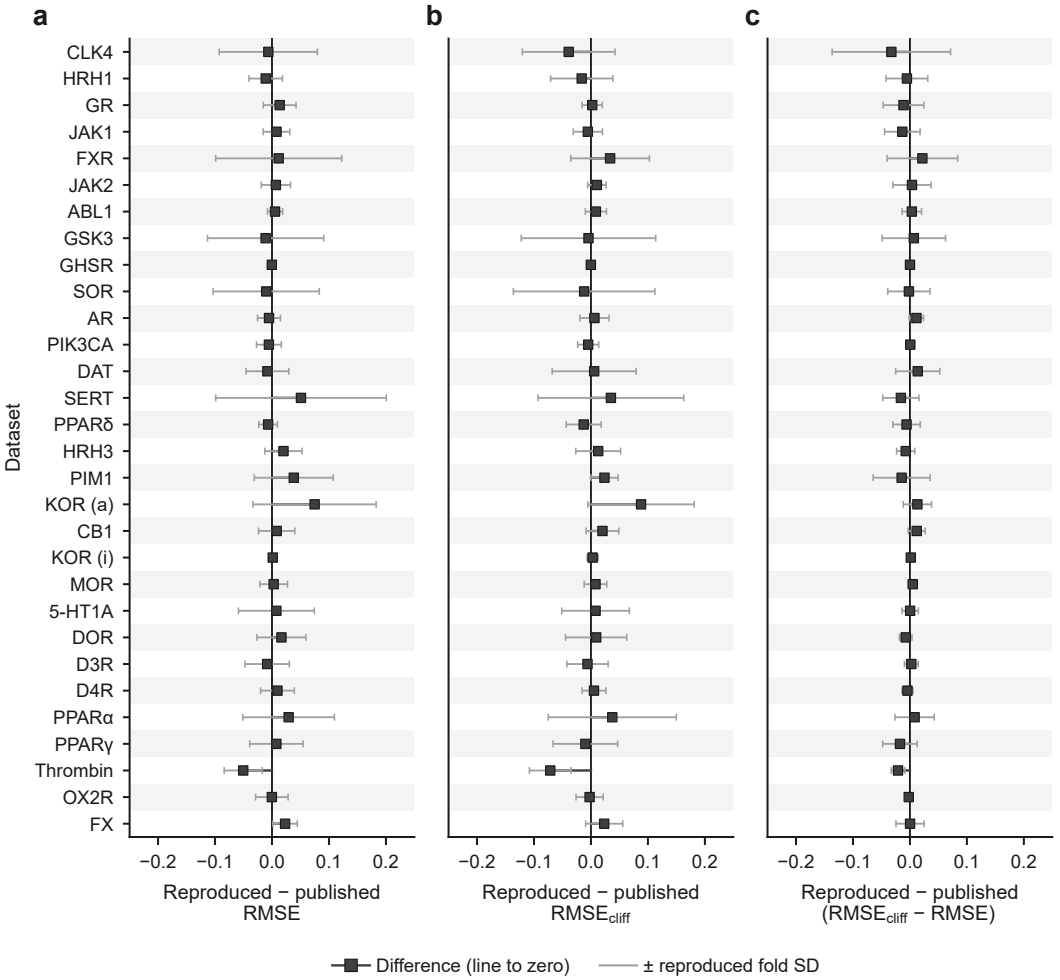



a
Reproduced RMSE
Published RMSE
1.4
1.3
1.2
1.1
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.4
0.6
0.8
1.0
1.2
1.4
Mean difference +0.007
Mean |difference| 0.015
Pearson r 0.98
n = 30 datasets
b
Reproduced RMSE$_{cliff}$
Published RMSE$_{cliff}$
1.3
1.2
1.1
1.0
0.9
0.8
0.7
0.7
0.8
0.9
1.0
1.1
1.2
1.3
Mean difference +0.005
Mean |difference| 0.017
Pearson r 0.97
n = 30 datasets
c
Reproduced (RMSE$_{cliff}$ − RMSE)
Published (RMSE$_{cliff}$ − RMSE)
0.5
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.2
0.0
0.2
0.4
Mean difference −0.002
Mean |difference| 0.009
Pearson r 0.99
n = 30 datasets
Reproduced GCN (this work)
Identity (equal scores)

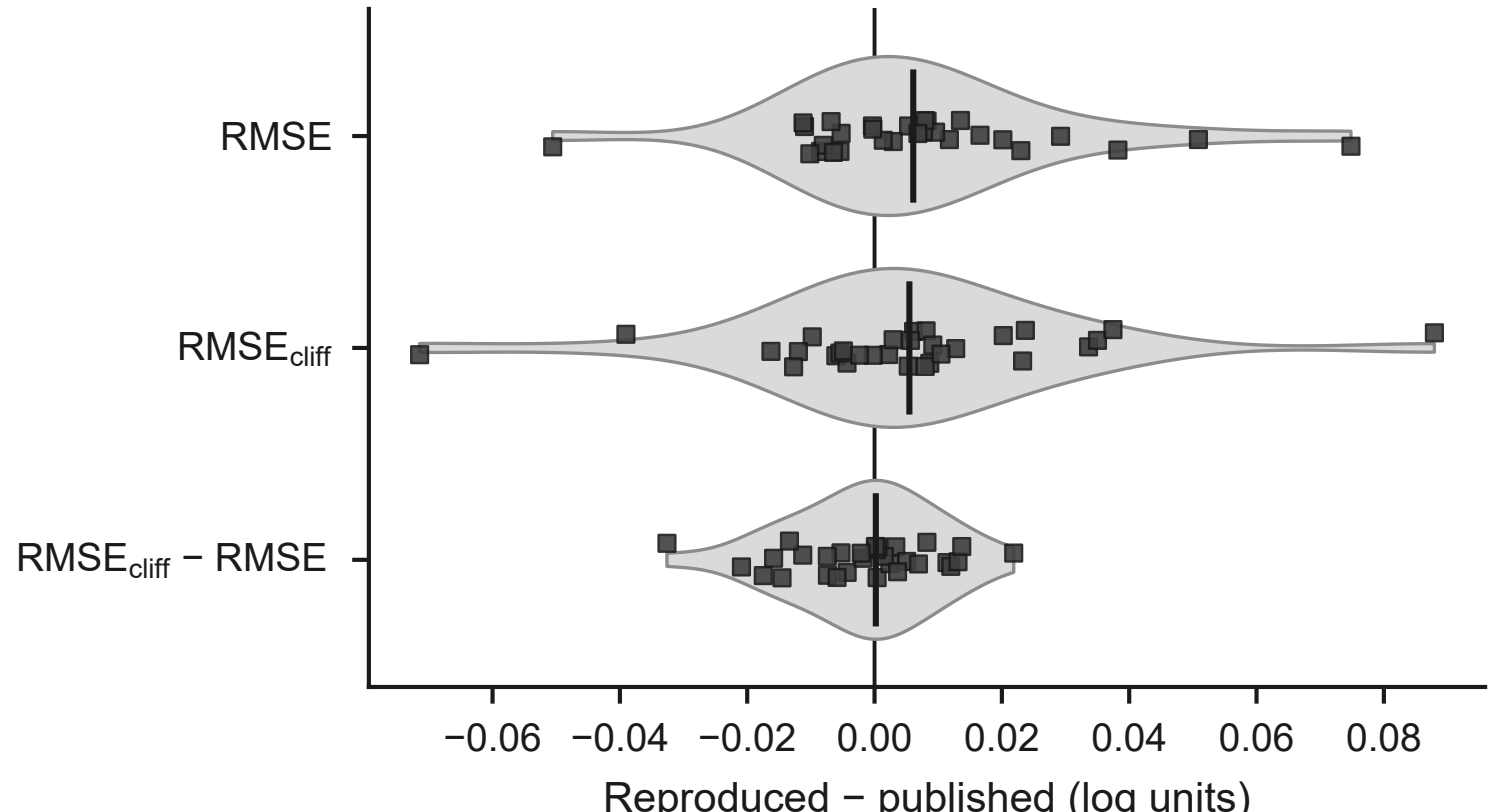

a
Reproduced − published RMSE
0.05
0.00
−0.05
+1.96 SD +0.050
Mean +0.007
−1.96 SD −0.036
0.6
0.8
1.0
1.2
Mean of published and reproduced RMSE


b
Reproduced − published RMSE$_{cliff}$
0.05
0.00
−0.05
+1.96 SD +0.057
Mean +0.005
−1.96 SD −0.047
0.8
0.9
1.0
1.1
1.2
Mean of published and reproduced RMSE$_{cliff}$

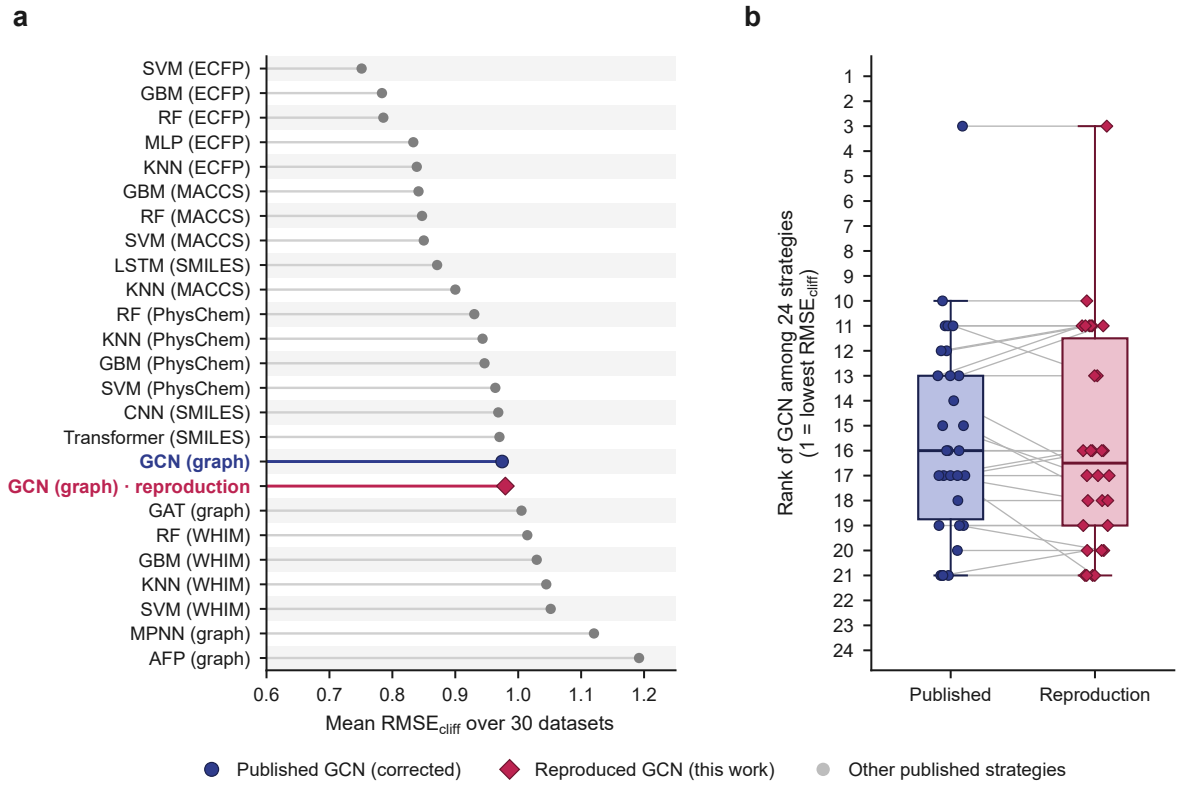
published_reproduced_differences_by_dataset — Reproduced minus published, per dataset

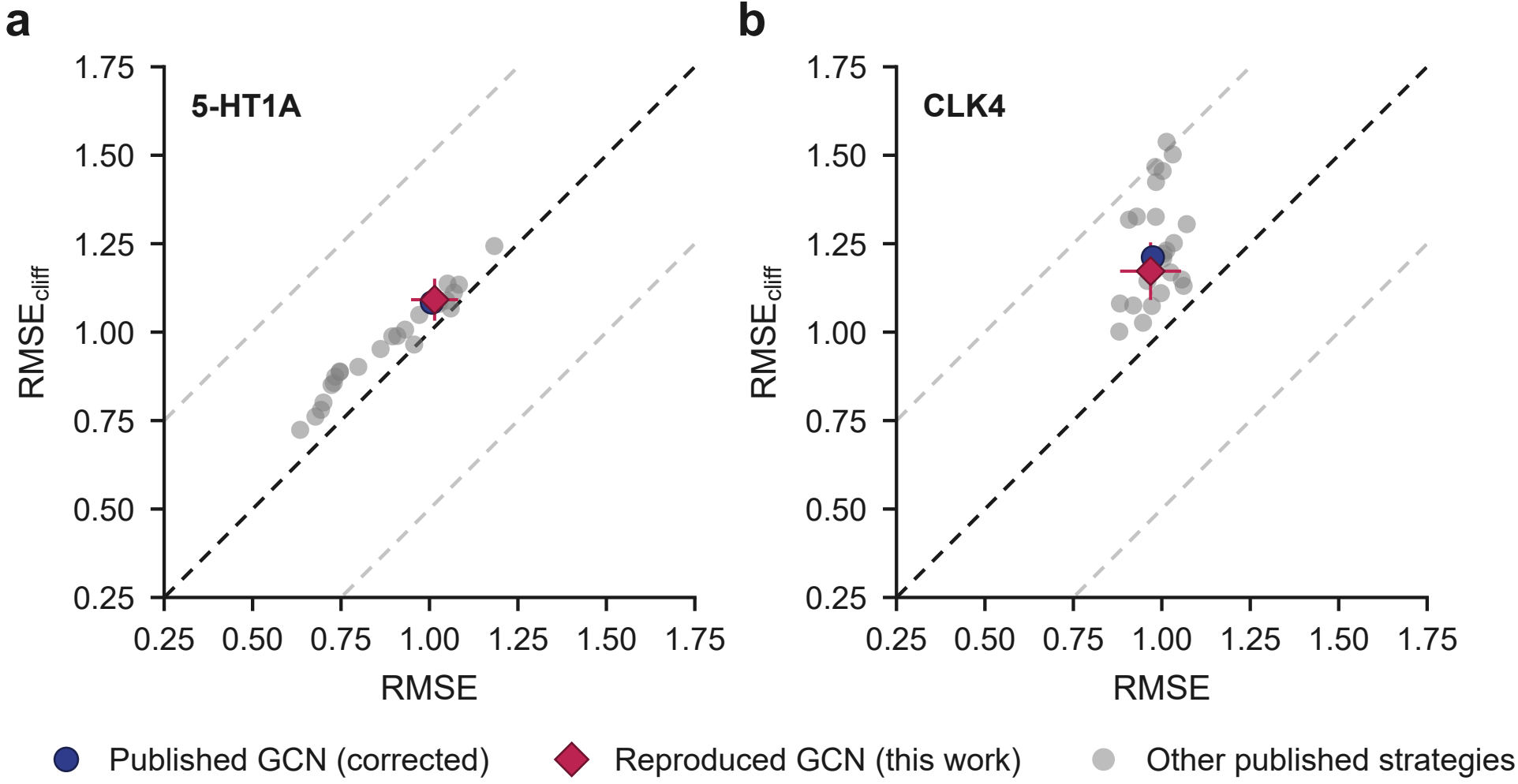


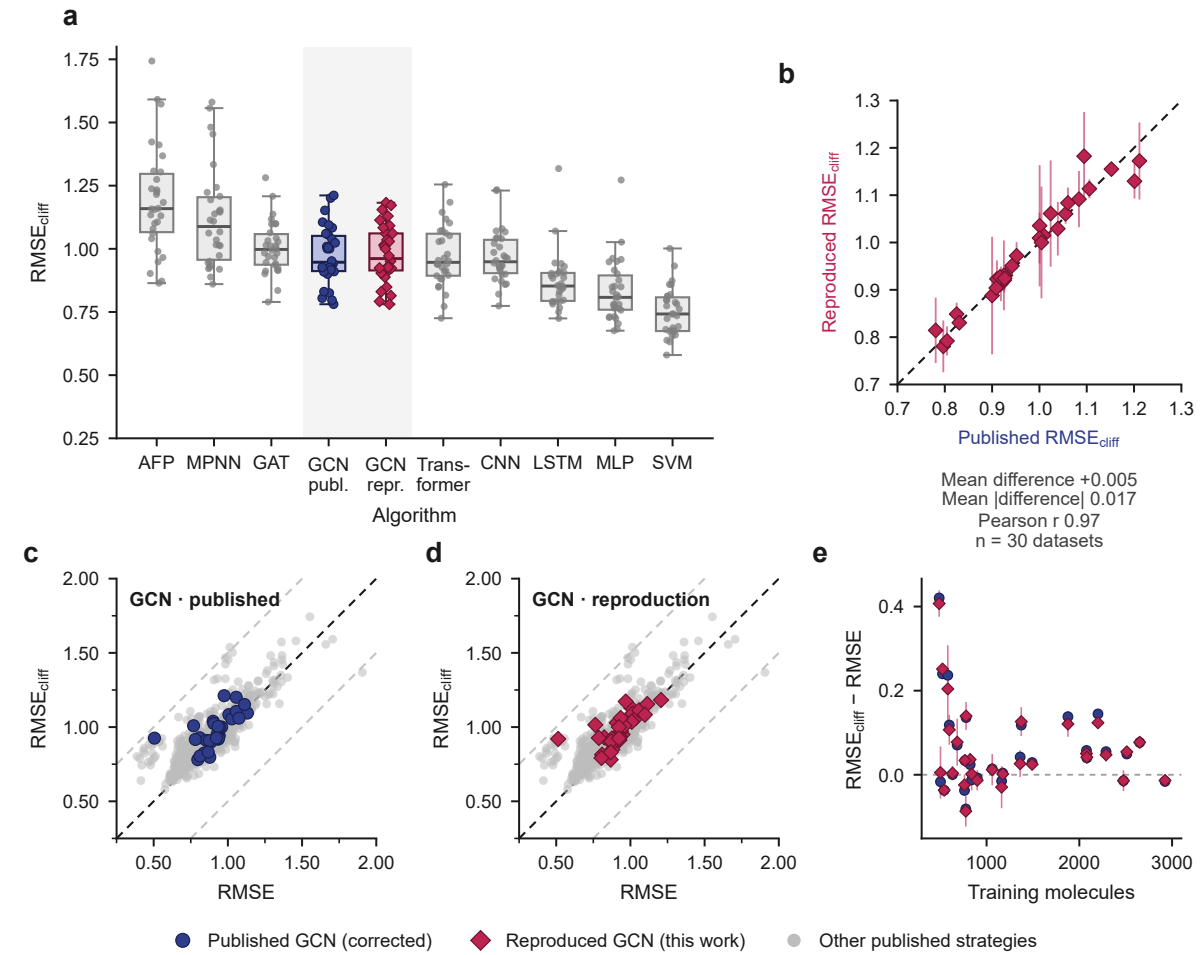
published_reproduced_difference_distribution — Distribution of reproduced minus published differences

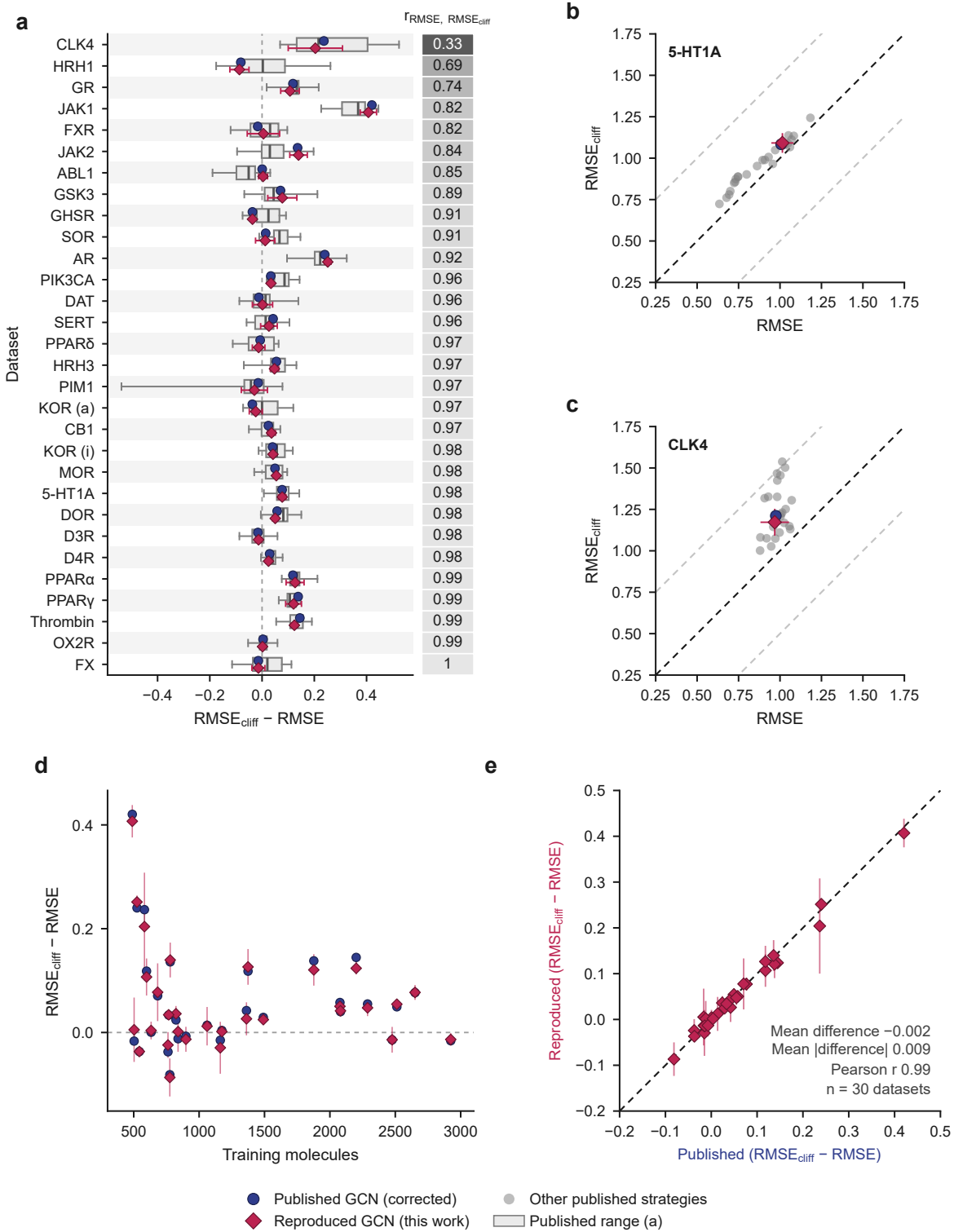


published_strategy_ranking_context — Where the reproduced GCN sits among published strategies

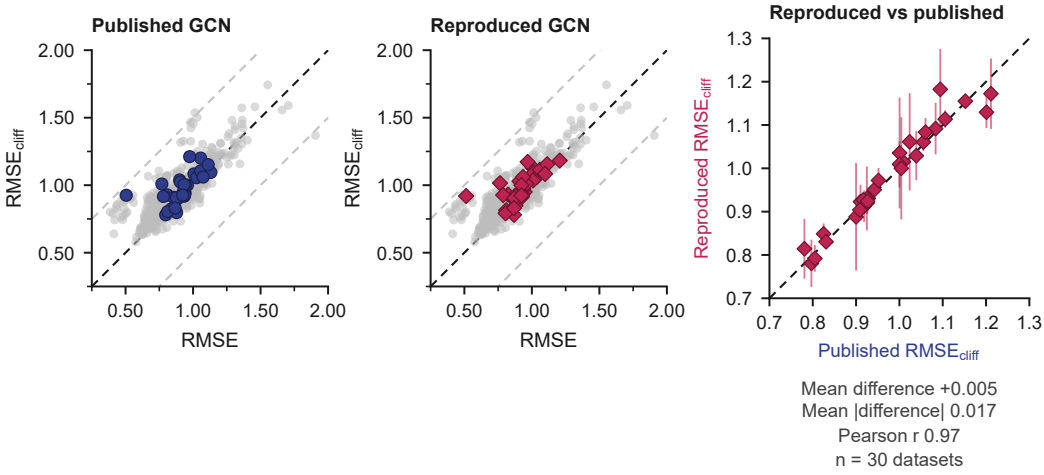



a
5-HT1A
b
CLK4
RMSE$_{cliff}$
RMSE
0.25
0.50
0.75
1.00
1.25
1.50
1.75
Published GCN (corrected)
Reproduced GCN (this work)
Other published strategies


a
RMSE_cliff
1.75
1.50
1.25
1.00
0.75
0.50
0.25
AFP
MPNN
GAT
GCN publ.
GCN repr.
Trans-former
CNN
LSTM
MLP
SVM
Algorithm
b
Reproduced RMSE_cliff
Published RMSE_cliff
1.3
1.2
1.1
1.0
0.9
0.8
0.7
Mean difference +0.005
Mean |difference| 0.017
Pearson r 0.97
n = 30 datasets
c
GCN · published
RMSE_cliff
RMSE
2.00
1.50
1.00
0.50
d
GCN · reproduction
RMSE_cliff
RMSE
e
RMSE_cliff − RMSE
0.4
0.2
0.0
1000
2000
3000
Training molecules
Published GCN (corrected)
Reproduced GCN (this work)
Other published strategies


a
$r_{RMSE,\ RMSE_{cliff}}$
CLK4 0.33
HRH1 0.69
GR 0.74
JAK1 0.82
FXR 0.82
JAK2 0.84
ABL1 0.85
GSK3 0.89
GHSR 0.91
SOR 0.91
AR 0.92
PIK3CA 0.96
DAT 0.96
SERT 0.96
PPARδ 0.97
HRH3 0.97
PIM1 0.97
KOR (a) 0.97
CB1 0.97
KOR (i) 0.98
MOR 0.98
5-HT1A 0.98
DOR 0.98
D3R 0.98
D4R 0.98
PPARα 0.99
PPARγ 0.99
Thrombin 0.99
OX2R 0.99
FX 1
Dataset
−0.4 −0.2 0.0 0.2 0.4
$RMSE_{cliff}$ − RMSE
b
5-HT1A
$RMSE_{cliff}$
RMSE
0.25 0.50 0.75 1.00 1.25 1.50 1.75
c
CLK4
$RMSE_{cliff}$
RMSE
0.25 0.50 0.75 1.00 1.25 1.50 1.75
d
$RMSE_{cliff}$ − RMSE
0.0 0.2 0.4
500 1000 1500 2000 2500 3000
Training molecules
e
Reproduced ($RMSE_{cliff}$ − RMSE)
Published ($RMSE_{cliff}$ − RMSE)
−0.2 −0.1 0.0 0.1 0.2 0.3 0.4 0.5
Mean difference −0.002
Mean |difference| 0.009
Pearson r 0.99
n = 30 datasets
Published GCN (corrected)
Reproduced GCN (this work)
Other published strategies
Published range (a)

gcn_results_overview — README summary banner



ChEMBL dataset	Target	n_{train}	n_{cliff}
CHEMBL2835_Ki	JAK1	489	13
CHEMBL2047_EC50	FXR	503	50
CHEMBL1871_Ki	AR	525	32
CHEMBL4616_EC50	GHSR	543	73
CHEMBL4203_Ki	CLK4	582	13
CHEMBL2034_Ki	GR	598	52
CHEMBL1862_Ki	ABL1	633	67
CHEMBL262_Ki	GSK3	683	32
CHEMBL237_EC50	KOR (a)	762	91
CHEMBL4005_Ki	PIK3CA	767	81
CHEMBL231_Ki	HRH1	776	48
CHEMBL2971_Ki	JAK2	779	34
CHEMBL218_EC50	CB1	823	76
CHEMBL238_Ki	DAT	839	55
CHEMBL3979_EC50	PPARδ	899	94
CHEMBL287_Ki	SOR	1061	103
CHEMBL2147_Ki	PIM1	1162	116
CHEMBL4792_Ki	OX2R	1174	160
CHEMBL228_Ki	SERT	1362	127
CHEMBL239_EC50	PPARα	1374	141
CHEMBL219_Ki	D4R	1491	148
CHEMBL235_EC50	PPARγ	1877	178
CHEMBL236_Ki	DOR	2077	204
CHEMBL237_Ki	KOR (i)	2081	223
CHEMBL204_Ki	Thrombin	2201	220
CHEMBL264_Ki	HRH3	2288	239
CHEMBL244_Ki	FX	2476	296
CHEMBL233_Ki	MOR	2512	259
CHEMBL214_Ki	5-HT1A	2651	245
CHEMBL234_Ki	D3R	2924	320

